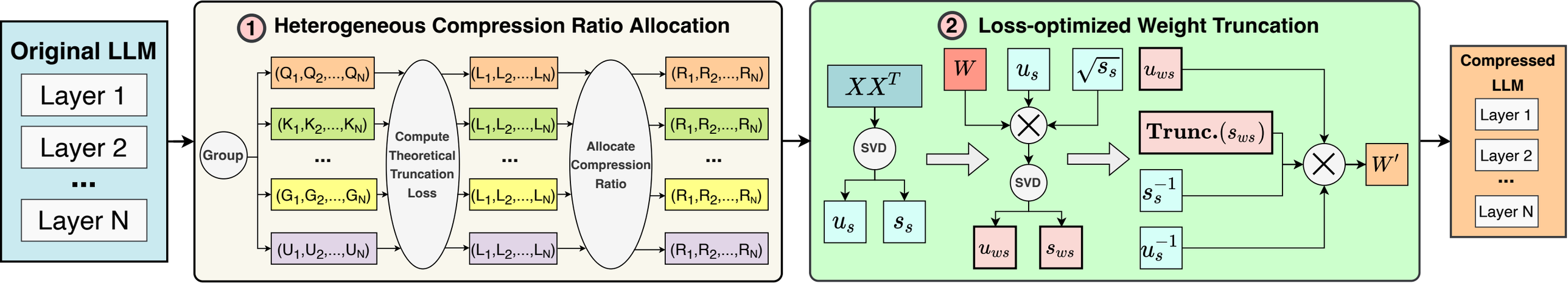
Original LLM
Layer 1
Layer 2
...
Layer N
1 Heterogeneous Compression Ratio Allocation
Group
(Q1,Q2,...,QN)
(K1,K2,...,KN)
...
(G1,G2,...,GN)
(U1,U2,...,UN)
Compute Theoretical Truncation Loss
(L1,L2,...,LN)
(L1,L2,...,LN)
...
(L1,L2,...,LN)
(L1,L2,...,LN)
Allocate Compression Ratio
(R1,R2,...,RN)
(R1,R2,...,RN)
...
(R1,R2,...,RN)
(R1,R2,...,RN)
2 Loss-optimized Weight Truncation
XX^T
SVD
u_s
s_s
W
u_s
$\sqrt{s_s}$
SVD
u_{ws}
s_{ws}
u_{ws}
Trunc.(s_{ws})
s_s^{-1}
u_s^{-1}
W'
Compressed LLM
Layer 1
Layer 2
...
Layer N